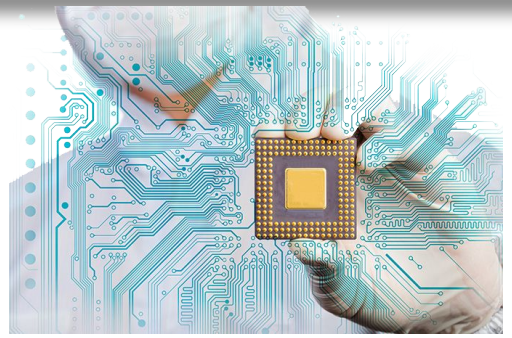
REPASO

SALCK Y SKEW

PRINCIPIO DE DISEÑO VLSI
DISEÑO DIGITAL CON TÉCNICAS DE ALTO NIVEL
ÁREA DISPOSITIVOS
DEPTO. DE ELECTRÓNICA Y COMPUTACIÓN
FI.UNMdP



REPASO

slack

SLACK DE SETUP

$$t_{clk} - t_{c \to Q} - t_{LC} - t_{setup} = SLACK_{setup}$$

SLACK DE HOLD

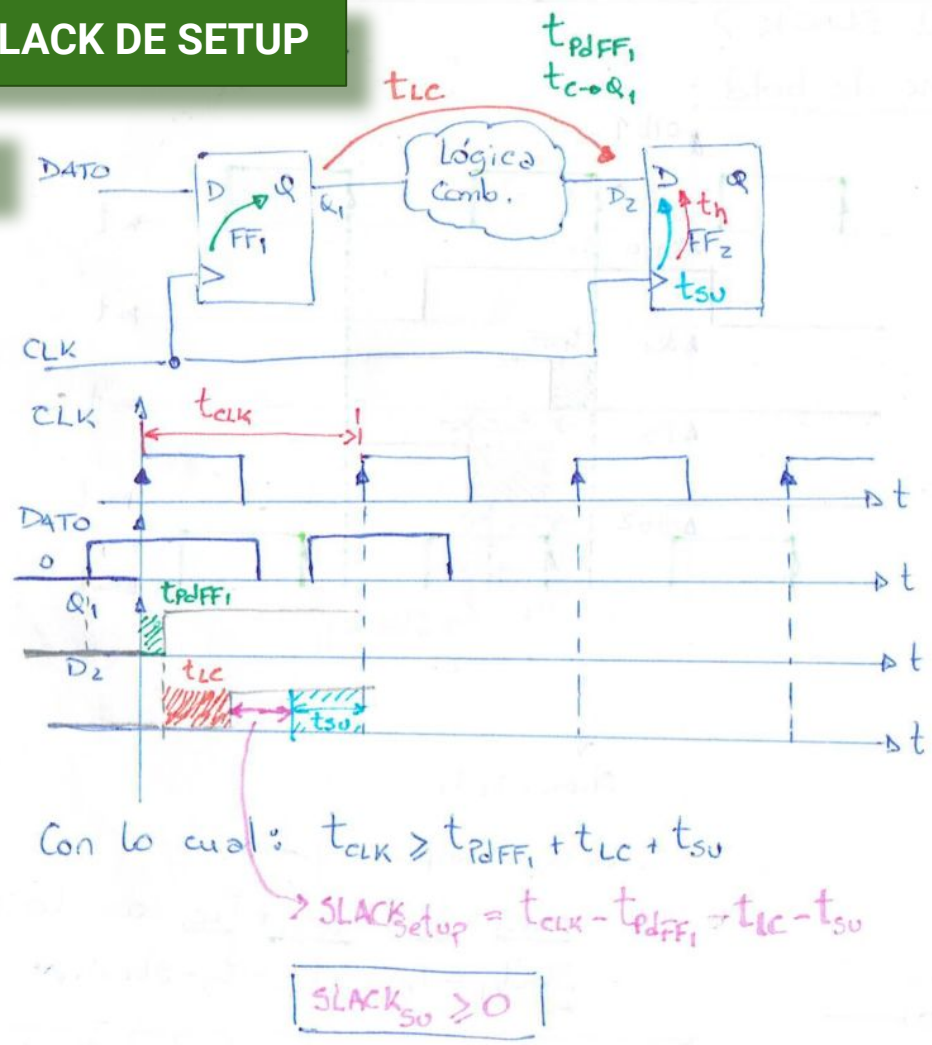
$$t_{c \to Q} + t_{LC} - t_{hold} = SLACK_{hold}$$

REPASO

slack

$$t_{clk} - t_{c \to Q} - t_{LC} - t_{setup} = SLACK_{setup}$$

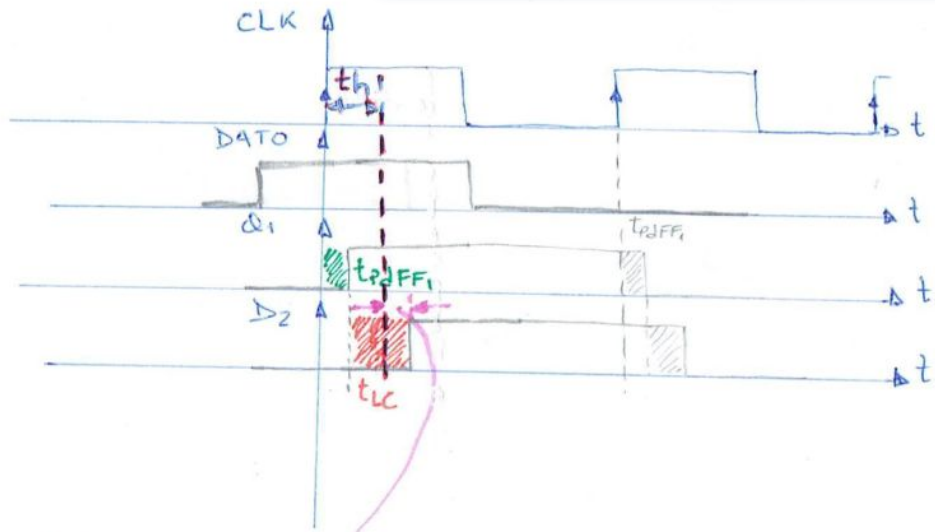
SLACK DE SETUP



REPASO

SLACK DE HOLD

$$t_{c\to Q} + t_{LC} - t_{hold} = SLACK_{hold}$$



Se debe cumplir: $t_h < t_{pdFF_1} + t_{LC}$

$SLACK_{hold} = t_{pdFF_1} + t_{LC} - t_h$

$\boxed{SLACK_h \geq 0}$

Si no se cumple $\Rightarrow$ HOLD VIOLATION

SLACK + SKEW

¿Cómo afecta el SKEW al SLACK?

$$SLACK_{setup} = t_{clk} - t_{c \to Q} - t_{LC} - t_{setup} + SKEW$$

$$SLACK_{hold} = t_{c \to Q} + t_{LC} - t_{hold} - SKEW$$

Si $SLACK_{su} < 0$ → **Setup Violation**

Si $SLACK_{h} < 0$ → **Hold Violation**